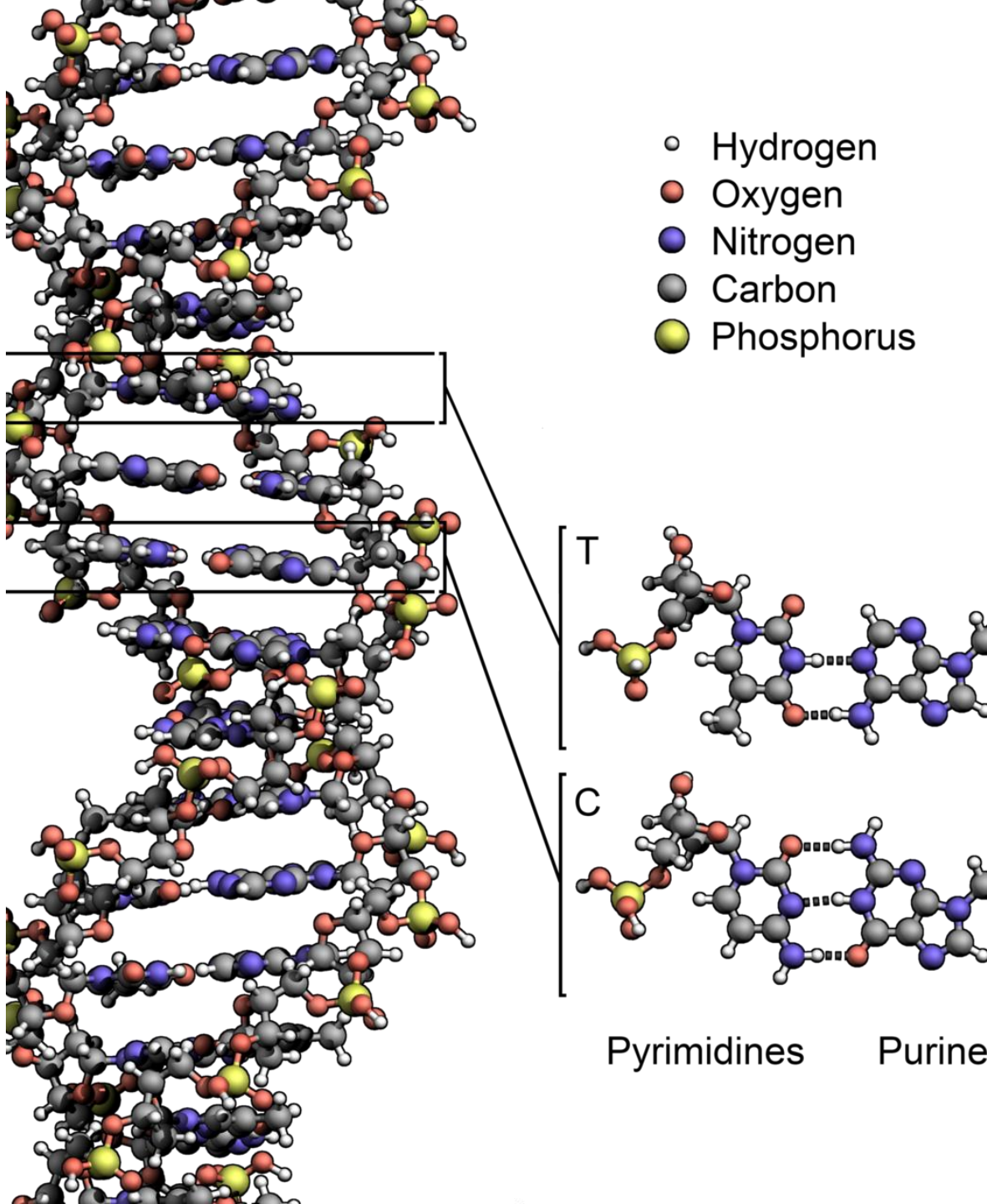
DP Biology – A1.2

Nucleic acids

- How do nucleic acids store information that can be inherited?
- Which features of DNA allow it to be copied accurately?

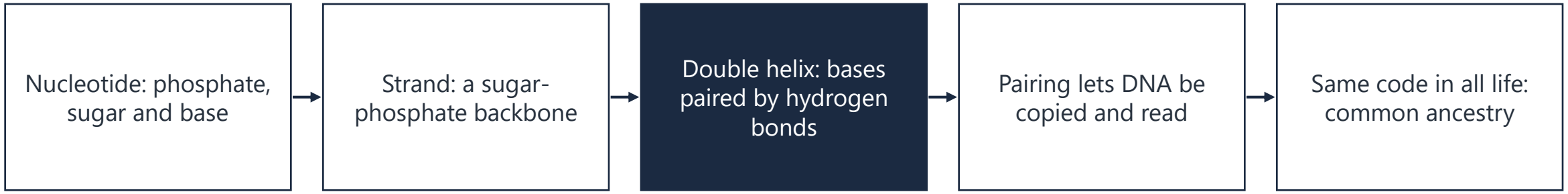
SL statements A1.2.1–10. Textbook pp. 14–24.
Image: structure of the DNA double helix. Zephyris, CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.2.1–5	Draw a nucleotide, and explain how nucleotides join into a strand with a sugar-phosphate backbone
2	A1.2.6–7	Draw a section of DNA, and compare the structures of DNA and RNA
3	A1.2.8–10	Explain base pairing, the capacity of DNA to store information, and the code as evidence of common ancestry

The big picture: from one nucleotide to inheritance



Nucleic acids are long chains of **nucleotides**. The way the chain is built gives it two jobs at once: a strong, unchanging backbone that holds the molecule together, and a varying sequence of **bases** that carries information.

In DNA, two strands are held together by pairs of bases that fit only one way. That is what lets cells copy DNA accurately and read it to make proteins.

Big question for this chapter

How can a molecule built from only four kinds of building block hold all the instructions for making a human?

Every living thing stores its genes as DNA

A1.2.1

Every living organism, from bacteria to blue whales, stores its genetic information as **DNA**. Cells use it to make proteins, and it is copied and passed on to offspring.

Some **viruses** use **RNA** instead of DNA as their genetic material. This does not break the rule, because viruses are not usually considered to be living: they have no metabolism of their own and can only reproduce inside a host cell.

DNA is found in the nucleus of eukaryotic cells (and in mitochondria and chloroplasts), and in the cytoplasm of prokaryotes.

Discussion

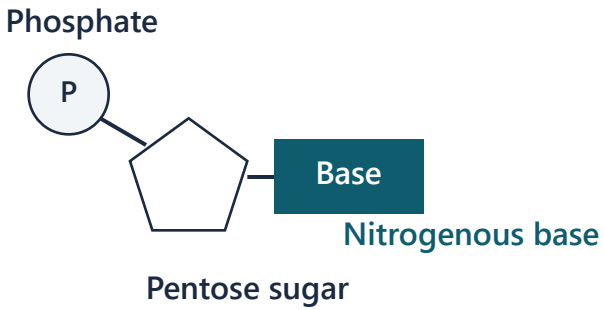
A coronavirus has RNA as its genetic material. Which characteristics of life does it lack? Does that make it non-living, or just a special case?

Link

A2.3 HL Viruses, and B2.2: where DNA is found in eukaryotic cells.

A nucleotide has three parts

A1.2.2



Draw circles, pentagons and rectangles in these positions. The phosphate and the base join different corners of the sugar.

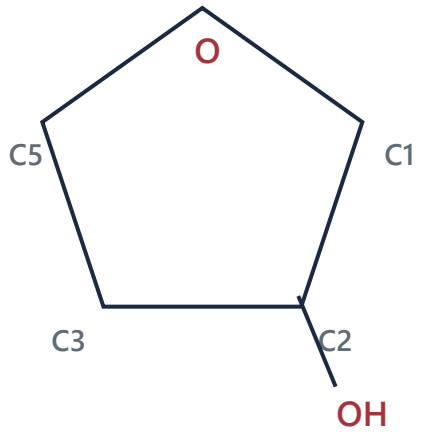
Part	What it is and does
Phosphate	An acidic group with a negative charge; links one nucleotide to the next
Pentose sugar	A five-carbon sugar: deoxyribose in DNA, ribose in RNA
Nitrogenous base	A ring containing nitrogen; one of four kinds in each nucleic acid; carries the information

Activity

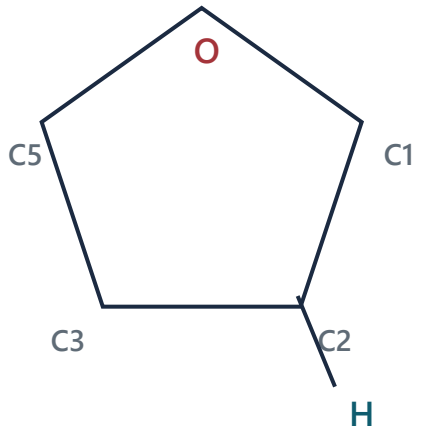
Draw one nucleotide using only these three shapes and label each part. Then draw three joined into a chain, as on slide 7.

Ribose and deoxyribose

A1.2.2



Ribose (RNA)



Deoxyribose (DNA)

Both sugars are **pentoses**: five carbon atoms, four of them in a ring with an oxygen.

The only difference is at **carbon 2**: ribose has an -OH group there, deoxyribose has just -H.

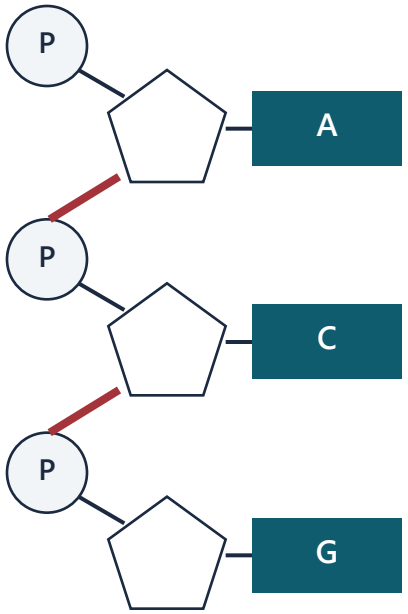
"Deoxy" means "missing an oxygen": deoxyribose has one oxygen atom fewer.

Exam tip

You may be asked to sketch the difference between ribose and deoxyribose. A pentagon with the ring oxygen, and -OH or -H on carbon 2, is enough.

The sugar-phosphate backbone

A1.2.3



Red: covalent bonds joining sugar to phosphate

Nucleotides join when the **phosphate** of one bonds to the **sugar** of the next. These are strong covalent bonds.

The result is a continuous chain of alternating sugars and phosphates: the **sugar-phosphate backbone**. It is the same all the way along, whatever the bases are.

The bases stick out to one side. Because the backbone does not depend on them, **any base can follow any other**, which is what allows any sequence to be stored.

Key idea

A strong backbone keeps the sequence of bases intact; the varying bases carry the information.

The bases: the part that varies

A1.2.4

Base	Type	In DNA	In RNA
Adenine (A)	Purine	Yes	Yes
Guanine (G)	Purine	Yes	Yes
Cytosine (C)	Pyrimidine	Yes	Yes
Thymine (T)	Pyrimidine	Yes	No
Uracil (U)	Pyrimidine	No	Yes

Each nucleic acid uses **four** bases. DNA uses A, C, G and T; RNA uses A, C, G and U.

Purines (A, G) have two rings; **pyrimidines** (C, T, U) have one.

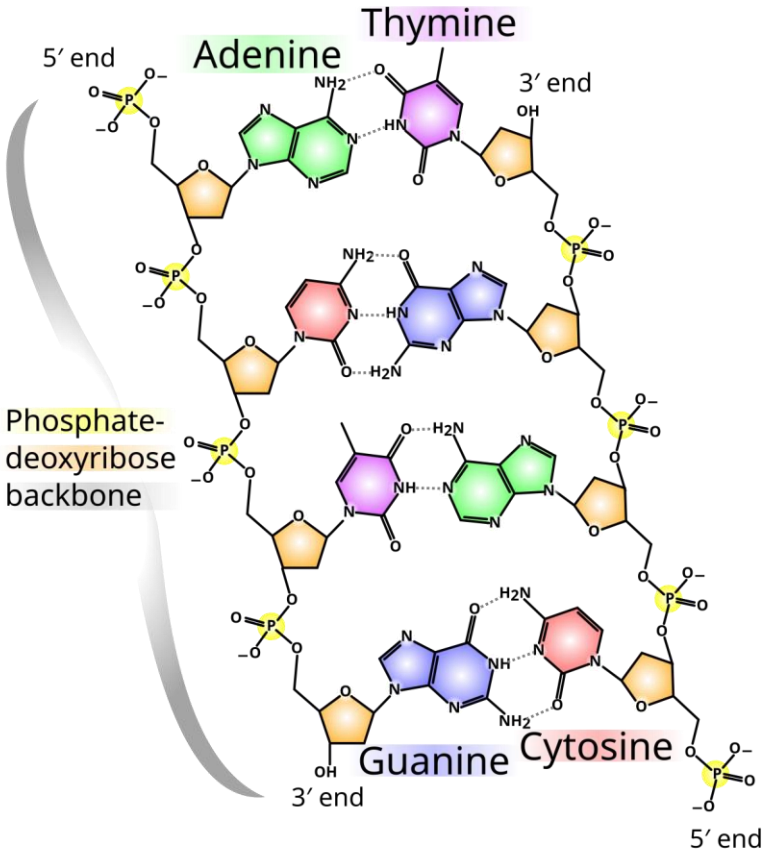
The bases are the only part that differs between nucleotides, so the **base sequence** is the information, like the order of letters in a word.

Memory aid

Purines are "pure as gold": Adenine and Guanine. Pyrimidines have the letter y: cytosine, thymine (and uracil).

DNA is a double helix

A1.2.6



DNA has **two strands** wound around each other to form a **double helix**.

The two strands are **antiparallel**: they run in opposite directions, so one is upside down compared with the other.

The strands are held together by **hydrogen bonds** between pairs of bases in the middle. **Adenine pairs with thymine**, and **guanine pairs with cytosine**: these are **complementary** base pairs.

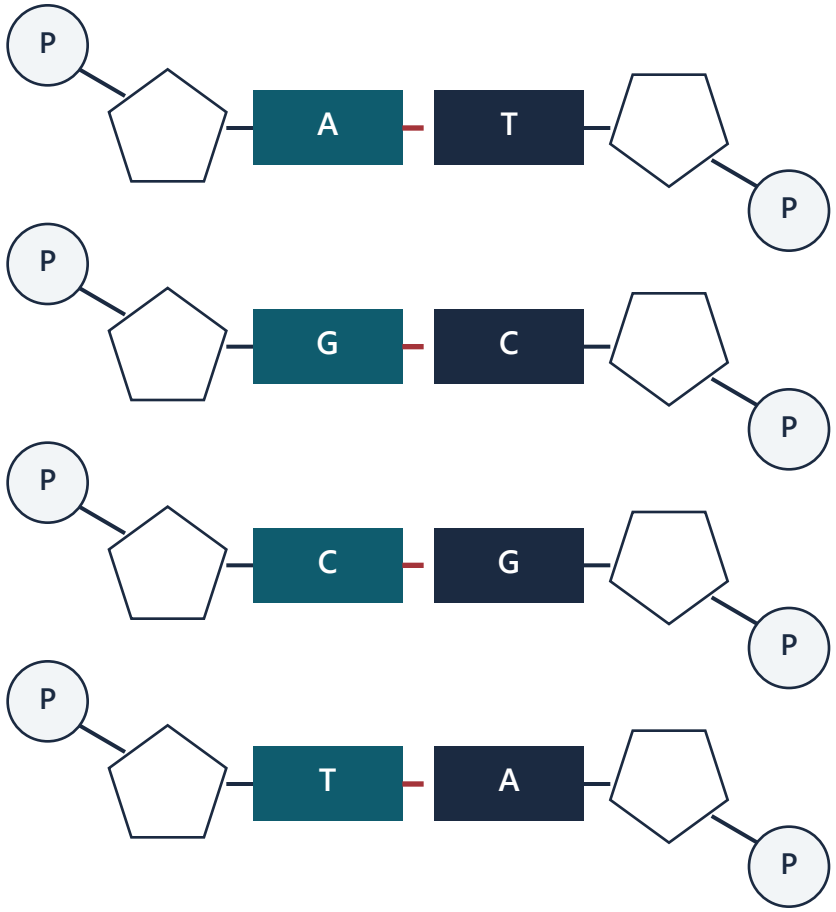
Why this shape works

The backbones are on the outside, protecting the bases inside. Each strand carries the information needed to rebuild the other.

The chemical structure of DNA. Madprime, CC BY-SA 3.0.

How to draw a section of DNA

A1.2.6



Dashed red: hydrogen bonds between bases

For full marks, a drawing of DNA should show:

1. Two strands of nucleotides, each with a **sugar-phosphate backbone**.
2. The strands **antiparallel**: the second strand drawn upside down.
3. Bases paired **A with T** and **G with C**.
4. **Hydrogen bonds** between the bases, and covalent bonds along each backbone, labelled.

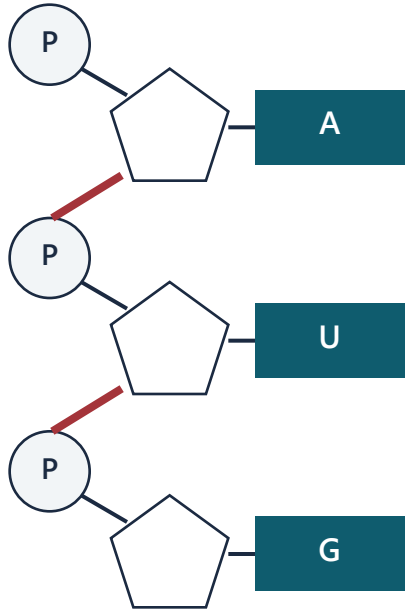
RNA: a polymer built by condensation

A1.2.5

RNA is a polymer: a long chain of nucleotide **monomers**.

Each new bond forms in a **condensation reaction**: the phosphate of one nucleotide joins the sugar of the next and a molecule of **water** is released.

RNA is usually a **single strand**. Its backbone is built in exactly the same way as a DNA strand, with the bases sticking out to one side.



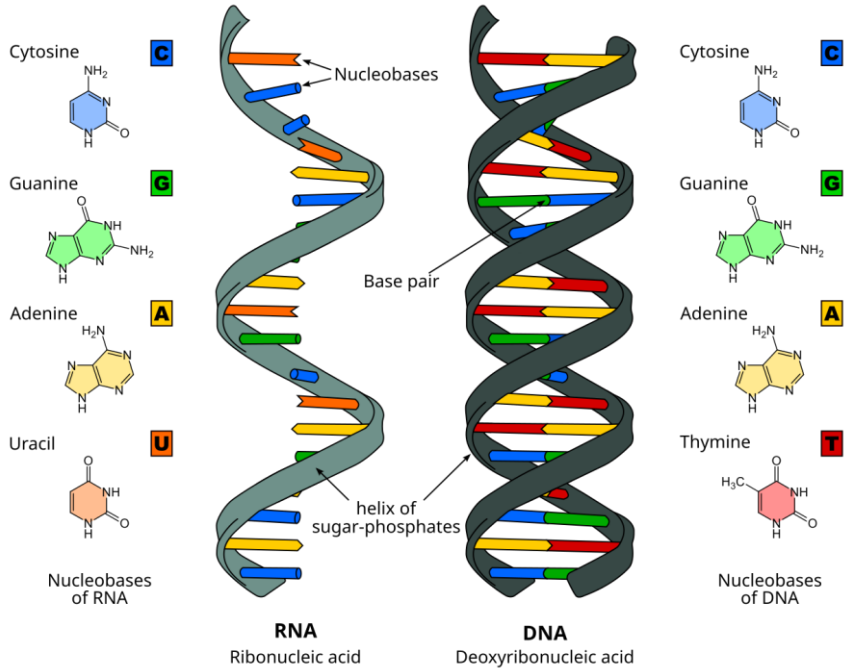
Schematic section of RNA: one strand

Link

B1.1 and B1.2: polysaccharides, lipids and proteins are also built by condensation and broken down by hydrolysis.

Comparing DNA and RNA

A1.2.7



Comparing RNA and DNA. Sponk, CC BY-SA 3.0.

Feature	DNA	RNA
Number of strands	Two (double helix)	One
Sugar	Deoxyribose	Ribose
Bases	A, C, G, T	A, C, G, U
Main role	Long-term store of genetic information	Copies of genes used to make proteins

Exam tip

Questions often ask for three differences. Strands, sugar and the base thymine or uracil are the three to learn first.

Check-in: building nucleic acids

A1.2.1–7

1. Name the three parts of a nucleotide.
2. State which part of a nucleotide carries the genetic information.
3. Describe how two nucleotides are joined in a strand.
4. Draw a section of DNA four base pairs long, labelling the backbone and hydrogen bonds.
5. State three differences between DNA and RNA.

How to use this

Answer without your notes, then compare with a partner.

Why bases pair the way they do

A1.2.8

A two-ring base (purine) always pairs with a one-ring base (pyrimidine), so every pair is the same width and the helix is regular.

Hydrogen bonds form only where the right atoms lie opposite each other: **A with T** (or **A with U** in RNA) and **G with C**. These are **complementary** base pairs.

Because of this, the sequence of one strand fixes the sequence of the other: if one strand reads ATGC, the other must read TACG.

Strand 1	Strand 2
A	T
T	A
G	C
C	G
?	?

Try it

One strand of DNA reads AATGCCGT. Write the complementary strand, then the RNA that would pair with the first strand.

Base pairing makes copying and reading possible

A1.2.8

Process	How complementary pairing is used
Replication	The two strands separate and each is a **template**: free nucleotides pair with the exposed bases, giving two identical DNA molecules
Transcription	One strand of a gene is a template for RNA: nucleotides pair with it, with U pairing with A
Translation	Each three-base **anticodon** on a tRNA pairs with the matching codon on the mRNA, bringing the right amino acid

Key idea

Pairing is why genetic information can be copied with very few errors and read in the same way every time.

An almost limitless capacity to store information

A1.2.9

Length	Calculation	Sequences
1 base	4	4
2 bases	4 × 4	16
3 bases	4^3	64
10 bases	4 to the power 10	over 1 million

With four bases and **any** base allowed at each position, the number of possible sequences is **4 to the power n** for a length of n bases.

A human cell has about **3 billion** base pairs in each set of chromosomes, so the number of possible sequences is unimaginably large.

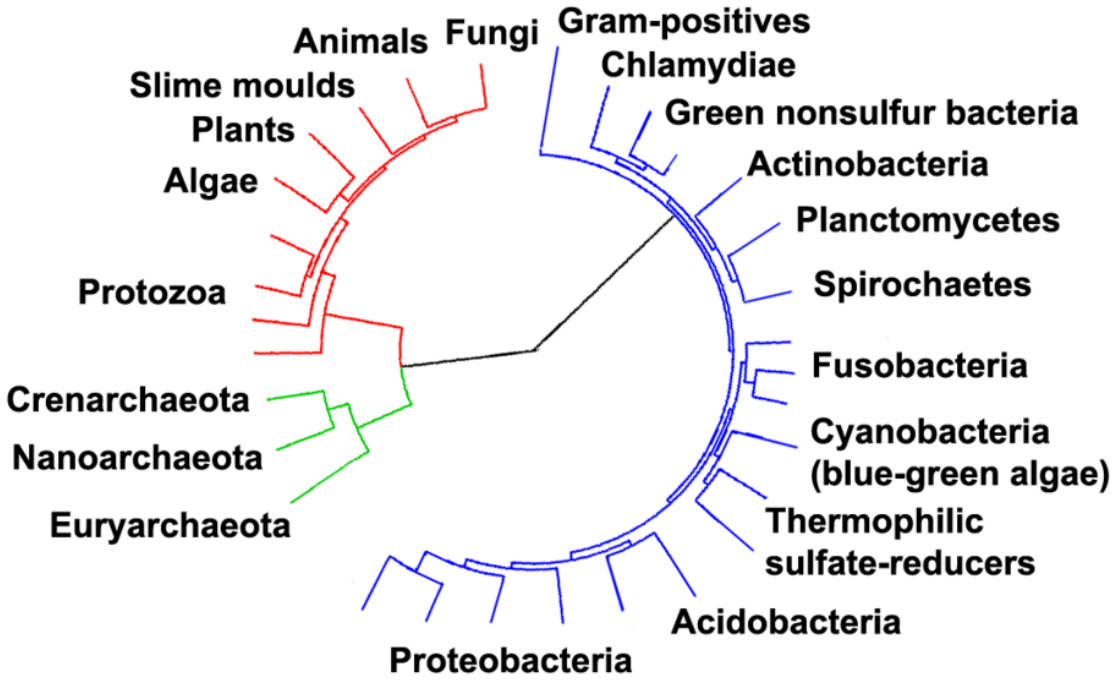
DNA is also very compact: a few grams could, in principle, hold all the data stored on the internet.

Discussion

Scientists have stored books, photos and films in DNA. Suggest why DNA might be better than a hard drive for storing data for thousands of years.

One genetic code for (almost) all of life

A1.2.10



A tree of life built from gene sequences. TimVickers, public domain.

The **genetic code** links each three-base codon to an amino acid. Almost every organism uses the **same code**.

It is very unlikely that the same code arose separately many times. The best explanation is that all life shares a **common ancestor** that already used it.

A few organisms and mitochondria have small variations, but these are rare and do not need to be learned.

Nature of science

This is an argument from evidence: a shared, arbitrary code is hard to explain in any other way.

Evidence: genes work in other species

A1.2.10



A firefly (Photuris). CC0.

When the gene for the light-producing enzyme of a **firefly** is put into a **tobacco plant**, the plant glows.

The plant's cells read the firefly gene and make the same protein, because both organisms use the same genetic code.

The same principle lets bacteria make **human insulin** from a human gene, which is how most insulin for diabetes is produced today.

Link

D1.3 and D1.2: gene transfer and protein synthesis. B2.2: why bacteria need no nucleus to make proteins from a human gene.

Working with data: base composition of DNA

A1.2.8

The table shows the percentage of each base in DNA from four organisms (published values, rounded).

Source of DNA	A / %	T / %	G / %	C / %
Human	30.9	29.4	19.9	19.8
Wheat	27.3	27.1	22.7	22.8
Yeast	31.3	32.9	18.7	17.1
E. coli	24.7	23.6	26.0	25.7

1. Compare the amount of A with T, and G with C, in each organism.
2. Explain these patterns using the structure of DNA.
3. Suggest why the values are not exactly equal.
4. Another organism's DNA is 18% G. Predict its percentage of A.

Check-in: pairing, information and the code

A1.2.8–10

1. One DNA strand reads GGATCC. Write the complementary strand.
2. Explain why complementary base pairing is needed for DNA replication.
3. Calculate the number of different sequences possible for a section of DNA 5 bases long.
4. Explain why the genetic code is evidence for common ancestry.
5. Outline why a bacterium can make a human protein.

How to use this

Write full sentences, then mark with a partner using the teacher’s answers.

Common mistakes to avoid

Not this	But this
The bases form the backbone of DNA.	The backbone is sugar and phosphate; the bases stick out from it.
A pairs with G, and C with T.	A pairs with T (or U in RNA), and G pairs with C.
Hydrogen bonds join the nucleotides in a strand.	Covalent bonds join them; hydrogen bonds join the two strands.
RNA contains thymine.	RNA contains uracil instead of thymine.
The two strands of DNA run in the same direction.	They are antiparallel: they run in opposite directions.

Check that you can...

1. **State** that every living organism stores its genes as DNA
2. **Draw** a nucleotide using simple shapes, and **sketch** ribose and deoxyribose
3. **Describe** how nucleotides form a sugar-phosphate backbone by condensation
4. **Draw** a section of DNA with antiparallel strands and complementary base pairs
5. **Compare** DNA and RNA
6. **Explain** how complementary base pairing is used when DNA is copied and when genes are expressed
7. **Calculate** the number of possible base sequences, and **explain** the storage capacity of DNA
8. **Explain** how the genetic code is evidence of common ancestry

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. List the three parts of a nucleotide.

2. State three differences between DNA and RNA.

3. A section of DNA is 4 bases long. Calculate how many sequences are possible.

Image credits

DNA Structure+Key+Labelled.pn NoBB.png – Zephyris – CC BY-SA 3.0

DNA chemical structure.svg – Madprime (talk · contribs) – CC BY-SA 3.0

Difference DNA RNA-EN.svg – File:Difference DNA RNA-DE.svg: Sponk / translation: Sponk – CC BY-SA 3.0

Collapsed tree labels simplified.png – TimVickers at English Wikipedia – Public domain

The Photuris firefly.jpg – Goodymeraj – CC0

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses